Supplementary Material for "Geometric Algebra-Based Optimal Control with Analytical Derivatives for Robotic Motion Planning"

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Abstract

This document provides detailed information on formulating the optimal control problem with analytical derivatives based on Geometric Algebra (GA). We present the differential action model and cost model constructed using GA, accompanied by their detailed derivation processes. Finally, benchmark results evaluated across various hardware platforms are provided.

1 Notation and Preliminaries

List symbols, dimensions, coordinate frames, assumptions, and conventions here.

2 Geometric Algebra Identities and Properties

2.1 Geometric Algebra

The core concept of GA is its associative bilinear operator called the geometric product. The geometric product of two vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$ is defined as the sum of the inner product, denoted as $\cdot$ and the outer product, denoted as $\wedge$ of the two vectors

$$\boldsymbol{ab} = \boldsymbol{a} \cdot \boldsymbol{b} + \boldsymbol{a} \wedge \boldsymbol{b}, \quad \forall \boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n \tag{1}$$

The inner product in the equation is a symmetric bilinear form describing the metric property of the space. It can be characterized by a signature (p, q, r) and a basis $\{\boldsymbol{e}_i\}$ of $\mathbb{R}^n$ such that

$$\begin{aligned} &\boldsymbol{e}_i \cdot \boldsymbol{e}_j = 0, \quad (i \neq j) \\ &\boldsymbol{e}_i \cdot \boldsymbol{e}_i = \begin{cases} 1 & 1 \leq i \leq p \\ -1 & p+1 \leq i \leq p+q \\ 0 & p+q+1 \leq i \leq n = p+q+r \end{cases} \end{aligned} \tag{2}$$

The outer product in (1) is the antisymmetric part of the tensor product. The result of the outer product of two vectors is an element in a new vector space, denoted as $\bigwedge^2(\mathbb{R}^n)$. The new vector space is defined as the space of bivectors, also named as the space with grade 2. Correspondingly, the basis of $\bigwedge^2(\mathbb{R}^n)$ is induced from the basis of $\mathbb{R}^n$, denoted as $\{\boldsymbol{e}_{ij} := \boldsymbol{e}_i \wedge \boldsymbol{e}_j, \text{ for } i \neq j (i, j = 1, 2, \ldots, n)\}$. Successively, the vector space with higher grades can be derived, denoted as $\bigwedge^3(\mathbb{R}^n), \ldots, \bigwedge^n(\mathbb{R}^n)$. It then naturally turns out that the geometric

product extends the vector space $\mathbb{R}^n$ to the direct sum of the linear space with different grades

$$\mathbb{G}_{p,q,r} = \bigwedge \mathbb{R}^n := \mathbb{R} \oplus \mathbb{R}^n \oplus \bigwedge^2(\mathbb{R}^n) \oplus \cdots \oplus \bigwedge^n(\mathbb{R}^n) \tag{3}$$

An element in the space is called a multivector. The multivector containing only the grade n part is called a pseudoscalar for the space. It is customary to normalize this to a unit pseudoscalar, denoted as $\mathbf{I}_n$. It turns out that the k-vector space is dual to the $(n-k)$-vector space. The duality is called the Poincaré Duality, and its explicit form can be defined as a linear operator

$$\begin{aligned} \mathcal{J} : \bigwedge^k(\mathbb{R}^n) &\longrightarrow \bigwedge^{n-k}(\mathbb{R}^n) \\ \text{s.t. } \mathcal{J}(\boldsymbol{e}_i^k) = \boldsymbol{e}_i^{n-k} &\Longleftrightarrow \boldsymbol{e}_i^k \wedge \boldsymbol{e}_i^{n-k} = \pm \mathbf{I}_n \end{aligned} \tag{4}$$

The regressive product is the dual of the exterior product (respectively corresponding to the "meet" and "join" in the following text).

$$\boldsymbol{a} \vee \boldsymbol{b} := \mathcal{J}(\mathcal{J}(\boldsymbol{a}) \wedge \mathcal{J}(\boldsymbol{b})), \quad \boldsymbol{a}, \boldsymbol{b} \in \mathbb{G}_{p,q,r} \tag{5}$$

The two most widely used variants of GA in robotics are 3D Conformal Geometric Algebra (CGA) $\mathbb{G}_{4,1,0}$ and 3D Plane-based (or Projective) Geometric Algebra (PGA) $\mathbb{G}_{3,0,1}$. In the subsequent sections of this paper, unless stated otherwise, we will adopt the 3D-PGA formulation, as many derivations for 3D-CGA are analogous. The sixteen basis vectors in $\mathbb{G}_{3,0,1}$ are

$$\begin{aligned} \{1, \boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_{01}, \boldsymbol{e}_{02}, \boldsymbol{e}_{03}, \\ \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}, \boldsymbol{e}_{123}, \boldsymbol{e}_{023}, \boldsymbol{e}_{031}, \boldsymbol{e}_{012}, \mathbf{I}\} \end{aligned} \tag{6}$$

Planes $\boldsymbol{p}$, lines $\boldsymbol{L}$, and points $\boldsymbol{Q}$ have explicit expressions in PGA [1]. They can be represented as vectors, bivectors and trivectors respectively in the corresponding projective space:

$$\begin{aligned} \text{Plane: } \boldsymbol{p} &= -p_d \boldsymbol{e}_0 + p_1 \boldsymbol{e}_1 + p_2 \boldsymbol{e}_2 + p_3 \boldsymbol{e}_3 \\ \text{Line: } \boldsymbol{L} &= m_1 \boldsymbol{e}_{01} + m_2 \boldsymbol{e}_{02} + m_3 \boldsymbol{e}_{03} + l_1 \boldsymbol{e}_{23} + l_2 \boldsymbol{e}_{31} + l_3 \boldsymbol{e}_{12} \\ \text{Point: } \boldsymbol{Q} &= \boldsymbol{e}_{123} - q_1 \boldsymbol{e}_{023} - q_2 \boldsymbol{e}_{031} - q_3 \boldsymbol{e}_{012} \end{aligned} \tag{7}$$

Any geometric element $\boldsymbol{X}$, such as a plane, line, or point, can have its rigid body motion described uniformly. Such motion can be achieved by twice reflections with two different planes. Suppose two different planes $\boldsymbol{a}$ and $\boldsymbol{b}$ are given, and the simplest rigid body motion of a multivector $\boldsymbol{X}$ in GA is then

$$\boldsymbol{M}_{\boldsymbol{ab}}(\boldsymbol{X}) = \boldsymbol{baXab} \tag{8}$$

The operation $\boldsymbol{M}_{\boldsymbol{ab}}$ is used to describe rigid body motion and is referred to as motors in GA. Define the motor as $\boldsymbol{M} = \boldsymbol{ab}$ with its reverse[1] $\widetilde{\boldsymbol{M}} = \boldsymbol{ba}$. The definition of $\boldsymbol{M}$ naturally satisfies $\widetilde{\boldsymbol{M}}\boldsymbol{M} = \boldsymbol{M}\widetilde{\boldsymbol{M}} = 1$. Therefore, the rigid body motion of a multivector $\boldsymbol{X}$ in GA is represented as:

$$\boldsymbol{X} = \boldsymbol{M}_{\boldsymbol{ab}}(\boldsymbol{X}) = \widetilde{\boldsymbol{M}}\boldsymbol{X}\boldsymbol{M} \tag{9}$$

2.2 Motor Group

Indeed, motors constitute a group, and it can be verified that they form a Lie group $\mathcal{M}$. Both the group multiplication operation (i.e., the geometric product in GA) and the inversion

[1] The reverse operator $\widetilde{\ }$ in GA denotes reversing the order of geometric products.

operation (i.e., the reverse operator in GA) are C^∞-smooth. It is worth noting that the motor group is isomorphic to unit dual quaternions. Both $\mathbb{G}_{4,1,0}$ and $\mathbb{G}_{3,0,1}$ contain the motor group $\mathcal{M}$ as a subalgebra, though with slightly different basis representations. The commonly used representations for these two algebras are provided below.

$$\begin{aligned}\mathcal{M}_{4,1,0} &\subset \operatorname{span}\{1, \boldsymbol{e}_{\infty 1}, \boldsymbol{e}_{\infty 2}, \boldsymbol{e}_{\infty 3}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}, \boldsymbol{e}_{\infty 123}\} \\ \mathcal{M}_{3,0,1} &\subset \operatorname{span}\{1, \boldsymbol{e}_{01}, \boldsymbol{e}_{02}, \boldsymbol{e}_{03}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}, \boldsymbol{e}_{0123}\}\end{aligned} \tag{10}$$

The Lie algebra $\mathbf{m}$ corresponding to the motor group is the bivector space.

$$\begin{aligned}\mathbf{m}_{4,1,0} &= \operatorname{span}\{\boldsymbol{e}_{\infty 1}, \boldsymbol{e}_{\infty 2}, \boldsymbol{e}_{\infty 3}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}\} \\ \mathbf{m}_{3,0,1} &= \operatorname{span}\{\boldsymbol{e}_{01}, \boldsymbol{e}_{02}, \boldsymbol{e}_{03}, \boldsymbol{e}_{23}, \boldsymbol{e}_{31}, \boldsymbol{e}_{12}\}\end{aligned} \tag{11}$$

The exponential map on the Lie group $\mathcal{M}$ is defined as follows:

$$\exp(\boldsymbol{B}) = e^{\boldsymbol{B}} = \sum_{k=0}^{\infty} \frac{\boldsymbol{B}^k}{k!}, \quad \forall \boldsymbol{B} \in \mathbf{m} \tag{12}$$

For any motor $\boldsymbol{M}$ in the group, it satisfies the property $\boldsymbol{M}\widetilde{\boldsymbol{M}} = 1$. Differentiating this identity yields $\dot{\boldsymbol{M}}\widetilde{\boldsymbol{M}} + \boldsymbol{M}\dot{\widetilde{\boldsymbol{M}}} = 0$. Consider a trajectory of the multivector $\boldsymbol{X}(t) = \widetilde{\boldsymbol{M}}\boldsymbol{X}^0\boldsymbol{M}$ parameterized with t. Taking derivatives on $\boldsymbol{X}(t)$, we have

$$\begin{aligned}\dot{\boldsymbol{X}}(t) &= \dot{\widetilde{\boldsymbol{M}}}\boldsymbol{X}^0\boldsymbol{M} + \widetilde{\boldsymbol{M}}\boldsymbol{X}^0\dot{\boldsymbol{M}} + \widetilde{\boldsymbol{M}}\dot{\boldsymbol{X}}^0\boldsymbol{M} \\ &= \dot{\widetilde{\boldsymbol{M}}}\boldsymbol{M}\boldsymbol{X} + \boldsymbol{X}\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}} + \widetilde{\boldsymbol{M}}\dot{\boldsymbol{X}}^0\boldsymbol{M} \\ &= -\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}}\boldsymbol{X} + \boldsymbol{X}\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}} + \widetilde{\boldsymbol{M}}\dot{\boldsymbol{X}}^0\boldsymbol{M} \\ &= \boldsymbol{X} \times (2\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}}) + \widetilde{\boldsymbol{M}}\dot{\boldsymbol{X}}^0\boldsymbol{M}\end{aligned} \tag{13}$$

Here, the commutator product $\times$ in the theory of GA is introduced

$$\boldsymbol{X}_1 \times \boldsymbol{X}_2 := \frac{1}{2}(\boldsymbol{X}_1\boldsymbol{X}_2 - \boldsymbol{X}_2\boldsymbol{X}_1), \forall \boldsymbol{X}_1, \boldsymbol{X}_2 \in \mathbb{G}_{p,q,r} \tag{14}$$

In the last line of Eq. (13), the term $2\widetilde{\boldsymbol{M}}\dot{\boldsymbol{M}}$ turns out to be the rigid body velocity with respect to the parameter t expressed in the spatial frame, denoted as $\boldsymbol{V}^s$. It is a bivector in the tangent space of $\mathcal{M}$ at the identity element 1. An important algebraic Lie bracket $[\cdot,\cdot]$ can be defined on the vector space $\mathbf{m}$. It is straightforward to verify, based on the properties of the geometric product, that this operator satisfies the Jacobi identity and other requisite conditions.

$$\begin{aligned}[\boldsymbol{B}_1, \boldsymbol{B}_2] &= \boldsymbol{B}_1\boldsymbol{B}_2 - \boldsymbol{B}_2\boldsymbol{B}_1 \\ &= 2\boldsymbol{B}_1 \times \boldsymbol{B}_2, \quad \forall \boldsymbol{B}_1, \boldsymbol{B}_2 \in \mathbf{m}\end{aligned} \tag{15}$$

For every $\boldsymbol{M} \in \mathcal{M}$, the adjoint action, $\mathrm{Ad}_{\boldsymbol{M}} : \mathbf{m} \to \mathbf{m}$, is a Lie algebra isomorphism that enables change of frames. Superscripts denote reference frames: s for spatial frame (or world frame), b for body frame.

$$\boldsymbol{V}^b = \mathrm{Ad}_{\boldsymbol{M}}(\boldsymbol{V}^s) = \boldsymbol{M}\boldsymbol{V}^s\widetilde{\boldsymbol{M}}, \forall \boldsymbol{M} \in \mathcal{M}, \boldsymbol{V}^s, \boldsymbol{V}^b \in \mathbf{m} \tag{16}$$

2.3 Kinematics

$$
\begin{aligned}
L_k &= \tilde{M}_{p(k)} L_k^0 M_{p(k)} \\
\dot{L}_k &= L_k \times V_k \\
\ddot{L}_k &= \dot{L}_k \times V_k + L_k \times \dot{V}_k
\end{aligned}
\tag{17}
$$

$$
\begin{aligned}
V_k &= \sum_{m \preceq k} \dot{q}_m L_m \\
\dot{V}_k &= \sum_{m \preceq k} \ddot{q}_m L_m + \dot{q}_m \dot{L}_m
\end{aligned}
\tag{18}
$$

2.4 Dynamics

According to Newtonian mechanics, momentum is the product of the mass and velocity of an object. In the PGA, a mass particle is modeled as a Euclidean point $\boldsymbol{Q}$ with a weight m. When the particle is attached to a rigid body, the velocity of $m\boldsymbol{Q}$ is $\dot{\boldsymbol{Q}}$, which equals $\boldsymbol{Q} \times \boldsymbol{V}$. The momentum of the mass particle is then modeled as the joining line of $m\boldsymbol{Q}$ and $\dot{\boldsymbol{Q}}$. It is modeled in the PGA as

$$
\boldsymbol{\Pi} = m\boldsymbol{Q} \vee \dot{\boldsymbol{Q}} = m\boldsymbol{Q} \vee (\boldsymbol{Q} \times \boldsymbol{V}) \tag{19}
$$

The force acting on the mass particle is modeled as the line determined by the 3D force vector $\boldsymbol{f}$ and the acting point $\boldsymbol{Q}$ [2]. It is named as a wrench and has the following form

$$
\boldsymbol{w} = \boldsymbol{Q} \vee \boldsymbol{f} \tag{20}
$$

Newton's second law, in modern form, states that the time derivative of the momentum is the force acting on the mass particle:

$$
\boldsymbol{w} = \dot{\boldsymbol{\Pi}} = m\boldsymbol{Q} \vee \ddot{\boldsymbol{Q}} \tag{21}
$$

For a rigid body, the momentum of the rigid body is defined as the integral of all mass particles contained in it:

$$
\boldsymbol{\Pi} = \int_{\Omega} [\rho \boldsymbol{Q} \vee (\boldsymbol{Q} \times \boldsymbol{V})] \, \mathrm{d}\boldsymbol{Q} \tag{22}
$$

Ω is the set containing all mass particles in the rigid body and ρ is mass density, a function about $\boldsymbol{Q}$. We recognize that the momentum is linear with respect to the velocity bivector $\boldsymbol{V}$. We define the total inertia map $\boldsymbol{I}(\cdot)$ as a linear function

$$
\boldsymbol{I}(\cdot) = \int_{\Omega} [\rho \boldsymbol{Q} \vee (\boldsymbol{Q} \times \cdot)] \, \mathrm{d}\boldsymbol{Q} \tag{23}
$$

The inertial duality map takes a bivector velocity $\boldsymbol{V}$ used to describe the kinematics of a motor, and uses the mass distribution of the rigid body to map it to a dual bivector $\boldsymbol{\Pi}$, representing a momentum that dynamical forces can act on. If $\boldsymbol{I}(\cdot)$ is expanded with respect to the basis of the bivector space, an inertia matrix similar to SVA and POE methods is obtained

$$
\underline{\boldsymbol{I}} = \begin{bmatrix} m\underline{\hat{\boldsymbol{c}}}^{\mathrm{T}} & m\underline{\boldsymbol{E}}_3 \\ \underline{\boldsymbol{J}} + m\underline{\hat{\boldsymbol{c}}}\underline{\hat{\boldsymbol{c}}}^{\mathrm{T}} & m\underline{\hat{\boldsymbol{c}}} \end{bmatrix} \tag{24}
$$

m represents the mass of the rigid body, $\underline{\boldsymbol{E}}_3$ is the 3×3 identity matrix, $\underline{\hat{\boldsymbol{c}}}$ is the skew-symmetric matrix representation generated from the position vector of the reference coordinate frame relative to the center of mass. $\underline{\boldsymbol{J}}$ is the inertia matrix with respect to the center-of-mass coordinate frame. These definitions are consistent with classical mechanics.

Suppose that the spatial inertia of the rigid body at the initial configuration is $\boldsymbol{I}(\cdot)$ and its coordinates w.r.t. the body-fixed frame $\underline{\boldsymbol{I}}_b$ is known. Then the momentum of the rigid body after a rigid motion $\boldsymbol{M}$ with the velocity $\boldsymbol{V}^s$ is

$$\begin{aligned}\boldsymbol{\Pi}^s &= \int_\Omega \left[\rho(\widetilde{\boldsymbol{M}}\boldsymbol{Q}^0\boldsymbol{M}) \vee ((\widetilde{\boldsymbol{M}}\boldsymbol{Q}^0\boldsymbol{M}) \times \boldsymbol{V}^s)\right] \mathrm{d}(\widetilde{\boldsymbol{M}}\boldsymbol{Q}^0\boldsymbol{M}) \\ &= \widetilde{\boldsymbol{M}}\left[\int_\Omega \rho\boldsymbol{Q}^0 \vee (\boldsymbol{Q}^0 \times (\boldsymbol{M}\boldsymbol{V}^s\widetilde{\boldsymbol{M}}))\mathrm{d}\boldsymbol{Q}^0\right]\boldsymbol{M} \\ &= \widetilde{\boldsymbol{M}}\boldsymbol{I}_b(\boldsymbol{M}\boldsymbol{V}^s\widetilde{\boldsymbol{M}})\boldsymbol{M} = \widetilde{\boldsymbol{M}}\boldsymbol{I}_b(\boldsymbol{V}^b)\boldsymbol{M}\end{aligned} \tag{25}$$

where $\boldsymbol{Q}^0$ is the initial position of $\boldsymbol{Q}$. Denote $\boldsymbol{I}_b(\boldsymbol{V}^b)$ as $\boldsymbol{\Pi}_b$, i.e., the momentum expressed in the body-fixed coordinate frame. Then equation (25) is simplified as

$$\boldsymbol{\Pi}^s = \widetilde{\boldsymbol{M}}\boldsymbol{\Pi}^b\boldsymbol{M} \tag{26}$$

The derivative of the momentum is formulated, then Newtonian dynamics is constructed with reference to Equation (21).

$$\begin{aligned}\boldsymbol{w}^s &= \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Pi}^s = \frac{\mathrm{d}}{\mathrm{d}t}(\widetilde{\boldsymbol{M}}\boldsymbol{I}_b(\boldsymbol{V}^b)\boldsymbol{M}) \\ &= (\widetilde{\boldsymbol{M}}\boldsymbol{I}_b(\boldsymbol{V}^b)\boldsymbol{M}) \times \boldsymbol{V}^s + \widetilde{\boldsymbol{M}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{I}_b(\boldsymbol{V}^b)\boldsymbol{M} \\ &= \widetilde{\boldsymbol{M}}\left[\boldsymbol{I}_b(\boldsymbol{V}^b) \times \boldsymbol{V}^b + \boldsymbol{I}_b(\dot{\boldsymbol{V}}^b)\right]\boldsymbol{M}\end{aligned} \tag{27}$$

where $\boldsymbol{w}^s$ is the total wrench in the spatial frame. In the body frame, since $\boldsymbol{w}^b = \boldsymbol{M}\boldsymbol{w}^s\widetilde{\boldsymbol{M}}$, a compact dynamic equation can be constructed.

$$\boldsymbol{w}^b = \dot{\boldsymbol{\Pi}}^b = \boldsymbol{I}_b(\dot{\boldsymbol{V}}^b) + \boldsymbol{I}_b(\boldsymbol{V}^b) \times \boldsymbol{V}^b \tag{28}$$

3 First Order Partial Derivatives of ID

3.1 Definition and Basic Properties

Define the covariant derivative with respect to q_i

$$\mathcal{D}_i X := \frac{\partial X}{\partial q_i} + L_i \times X \tag{29}$$

Then define the partial derivative operators with respect to $\dot{q}$ and $\ddot{q}$

$$\mathcal{E}_i X := \frac{\partial X}{\partial \dot{q}_i}, \quad \mathcal{F}_i X := \frac{\partial X}{\partial \ddot{q}_i} \tag{30}$$

Define another operator composed of three items, which is similar to the idea presented in [3].

$$\mathcal{B}_{k,Y}(X) := I_k(Y \times X) + I_k(X) \times Y + I_k(Y) \times X \tag{31}$$

Next, we present the basic properties of these operators: as given below

$$\frac{\partial I_k(X)}{\partial q_i} = I_k(L_i \times X) - L_i \times I_k(X) + I_k\left(\frac{\partial X}{\partial q_i}\right) \tag{32}$$

We get

$$\mathcal{D}_i(I_k(X)) = \frac{\partial I_k}{\partial q_i} + L_i \times I_k(X) = I_k(L_i \times X) + I_k\left(\frac{\partial X}{\partial q_i}\right) \tag{33}$$

Thus, the covariant derivative commutes with the inertia matrix mapping I for the case where $i \preceq k$.

$$\mathcal{D}_i(I_k(X)) = I_k(\mathcal{D}_i X) \tag{34}$$

$$\mathcal{E}_i(I_k(X)) = I_k(\mathcal{E}_i X), \quad \mathcal{F}_i(I_k(X)) = I_k(\mathcal{F}_i X) \tag{35}$$

$$\begin{aligned}\mathcal{D}_i(X \times Y) &= \frac{\partial(X \times Y)}{\partial q_i} + L_i \times (X \times Y) = \frac{\partial X}{\partial q_i} \times Y + X \times \frac{\partial Y}{\partial q_i} - Y \times (L_i \times X) - X \times (Y \times L_i) \\ &= \left(\frac{\partial X}{\partial q_i} \times Y + (L_i \times X) \times Y\right) + \left(X \times \frac{\partial Y}{\partial q_i} + X \times (L_i \times Y)\right) = \mathcal{D}_i X \times Y + X \times \mathcal{D}_i Y\end{aligned} \tag{36}$$

3.2 Identities

$$\mathcal{D}_i \boldsymbol{L}_k = 0 \tag{37}$$

$$\mathcal{D}_i \boldsymbol{V}_k = \begin{cases} \dot{L}_i & i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{38}$$

$$\mathcal{D}_i \boldsymbol{V}_k = \begin{cases} \dot{L}_i & i \preceq k \\ 0 & \text{otherwise} \end{cases} \tag{39}$$

4 Detailed Derivation of the First- and Second-Order Derivatives of $\dot{\Pi}_k$ and $\tau_k = L_k * \dot{\Pi}_k$

In this appendix we provide a complete derivation of the first- and second-order derivatives of

$$\dot{\Pi}_k = I_k(\dot{V}_k) + [I_k(V_k), V_k], \qquad \tau_k = L_k * \dot{\Pi}_k.$$

The symbol $*$ denotes a scalar pairing with the property that, for any Lie-algebra-valued quantity X,

$$\frac{\partial}{\partial q_i}(L_k * X) = L_k * \left(L_i \times X + \frac{\partial X}{\partial q_i}\right), \tag{40}$$

$$\frac{\partial}{\partial \dot{q}_i}(L_k * X) = L_k * \frac{\partial X}{\partial \dot{q}_i}, \tag{41}$$

$$\frac{\partial}{\partial \ddot{q}_i}(L_k * X) = L_k * \frac{\partial X}{\partial \ddot{q}_i}. \tag{42}$$

4.1 Assumptions and basic identities

We start from the identities

$$\dot{L}_k = [L_k, V_k], \tag{43}$$
$$\ddot{L}_k = [\dot{L}_k, V_k] + [L_k, \dot{V}_k], \tag{44}$$
$$\frac{\partial L_k}{\partial q_i} = [L_k, L_i], \tag{45}$$
$$\frac{\partial V_k}{\partial q_i} = [V_k - V_i, L_i], \tag{46}$$
$$\frac{\partial V_k}{\partial \dot{q}_i} = L_i, \tag{47}$$
$$\frac{\partial \dot{L}_k}{\partial q_i} = [L_k, \dot{L}_i] - [L_i, \dot{L}_k], \tag{48}$$
$$\frac{\partial \dot{L}_k}{\partial \dot{q}_i} = [L_k, L_i], \tag{49}$$
$$\frac{\partial \dot{V}_k}{\partial q_i} = [\dot{V}_k - \dot{V}_i, L_i] + [V_k - V_i, \dot{L}_i], \tag{50}$$
$$\frac{\partial \dot{V}_k}{\partial \dot{q}_i} = [V_k - 2V_i, L_i], \tag{51}$$
$$\frac{\partial \dot{V}_k}{\partial \ddot{q}_i} = L_i, \tag{52}$$
$$\frac{\partial \ddot{L}_k}{\partial q_i} = [L_k, \ddot{L}_i] - [L_i, \ddot{L}_k] + 2\,\dot{L}_k \times \dot{L}_i, \tag{53}$$
$$\frac{\partial \ddot{L}_k}{\partial \dot{q}_i} = 2(L_k \times \dot{L}_i - L_i \times \dot{L}_k), \tag{54}$$
$$\frac{\partial \ddot{L}_k}{\partial \ddot{q}_i} = L_k \times L_i, \tag{55}$$
$$\frac{\partial I_k(X)}{\partial q_i} = I_k(L_i \times X) - L_i \times I_k(X), \qquad (i \preceq k), \tag{56}$$
$$\dot{\mathrm{I}}_k = I_k(\dot{V}_k) + [I_k(V_k), V_k]. \tag{57}$$

We also use the Jacobi identity

$$x \times (y \times z) + z \times (x \times y) + y \times (z \times x) = 0.$$

4.2 Definition of the covariant derivative

For convenience, define the q_i-direction covariant derivative

$$\mathcal{D}_i X := L_i \times X + \frac{\partial X}{\partial q_i}. \tag{58}$$

The choice (58) is natural because, using (45),

$$\mathcal{D}_i L_k = L_i \times L_k + \frac{\partial L_k}{\partial q_i} = [L_i, L_k] + [L_k, L_i] = 0.$$

Hence,

$$\mathcal{D}_i L_k = 0. \tag{59}$$

Moreover, using (56),

$$\begin{aligned}\mathcal{D}_i(I_k(X)) &= L_i \times I_k(X) + \frac{\partial I_k(X)}{\partial q_i} \\ &= L_i \times I_k(X) + I_k(L_i \times X) - L_i \times I_k(X) \\ &= I_k(L_i \times X). \end{aligned} \tag{60}$$

If $X = X(q)$ also depends on q, then by linearity,

$$\begin{aligned}\mathcal{D}_i(I_k(X)) &= L_i \times I_k(X) + \frac{\partial}{\partial q_i} I_k(X) \\ &= L_i \times I_k(X) + \left(\frac{\partial I_k}{\partial q_i}\right)(X) + I_k\left(\frac{\partial X}{\partial q_i}\right) \\ &= L_i \times I_k(X) + I_k(L_i \times X) - L_i \times I_k(X) + I_k\left(\frac{\partial X}{\partial q_i}\right) \\ &= I_k\left(L_i \times X + \frac{\partial X}{\partial q_i}\right) \\ &= I_k(\mathcal{D}_i X). \end{aligned} \tag{61}$$

Therefore,

$$\boxed{\mathcal{D}_i(I_k(X)) = I_k(\mathcal{D}_i X).} \tag{62}$$

4.3 Preliminary identities

Lemma 1 (Covariant derivatives of the basic quantities)**.** *The following identities hold:*

$$\mathcal{D}_i L_k = 0, \tag{63}$$

$$\mathcal{D}_i V_k = \dot{L}_i, \tag{64}$$

$$\mathcal{D}_i \dot{L}_k = L_k \times \dot{L}_i, \tag{65}$$

$$\mathcal{D}_i \dot{V}_k = \ddot{L}_i + V_k \times \dot{L}_i, \tag{66}$$

$$\mathcal{D}_i \ddot{L}_k = L_k \times \ddot{L}_i + 2\,\dot{L}_k \times \dot{L}_i. \tag{67}$$

In particular, after replacing k by i where needed,

$$\mathcal{D}_j \dot{L}_i = L_i \times \dot{L}_j, \tag{68}$$

$$\mathcal{D}_j \ddot{L}_i = L_i \times \ddot{L}_j + 2\,\dot{L}_i \times \dot{L}_j. \tag{69}$$

Proof. Equation (63) was already shown above.

Using (46),

$$\begin{aligned}\mathcal{D}_i V_k &= L_i \times V_k + \frac{\partial V_k}{\partial q_i} \\ &= [L_i, V_k] + [V_k - V_i, L_i] \\ &= [L_i, V_k] + [V_k, L_i] - [V_i, L_i] \\ &= -[V_i, L_i] \\ &= [L_i, V_i] \\ &= \dot{L}_i, \end{aligned} \tag{70}$$

which proves (64).

Using (48),

$$\begin{aligned}\mathcal{D}_i \dot{L}_k &= L_i \times \dot{L}_k + \frac{\partial \dot{L}_k}{\partial q_i} \\ &= [L_i, \dot{L}_k] + [L_k, \dot{L}_i] - [L_i, \dot{L}_k] \\ &= [L_k, \dot{L}_i] \\ &= L_k \times \dot{L}_i, \end{aligned} \tag{71}$$

which proves (65).

Using (50),

$$\begin{aligned}\mathcal{D}_i \dot{V}_k &= L_i \times \dot{V}_k + \frac{\partial \dot{V}_k}{\partial q_i} \\ &= [L_i, \dot{V}_k] + [\dot{V}_k - \dot{V}_i, L_i] + [V_k - V_i, \dot{L}_i] \\ &= [L_i, \dot{V}_k] + [\dot{V}_k, L_i] - [\dot{V}_i, L_i] + [V_k - V_i, \dot{L}_i] \\ &= [L_i, \dot{V}_i] + [V_k - V_i, \dot{L}_i]. \end{aligned} \tag{72}$$

Now use (44) with index i:

$$\ddot{L}_i = [\dot{L}_i, V_i] + [L_i, \dot{V}_i].$$

Hence

$$[L_i, \dot{V}_i] = \ddot{L}_i - [\dot{L}_i, V_i].$$

Substituting this into (72),

$$\begin{aligned}\mathcal{D}_i \dot{V}_k &= \ddot{L}_i - [\dot{L}_i, V_i] + [V_k - V_i, \dot{L}_i] \\ &= \ddot{L}_i + [V_i, \dot{L}_i] + [V_k, \dot{L}_i] - [V_i, \dot{L}_i] \\ &= \ddot{L}_i + [V_k, \dot{L}_i] \\ &= \ddot{L}_i + V_k \times \dot{L}_i, \end{aligned} \tag{73}$$

which proves (66).

Finally, using (53),

$$\begin{aligned}\mathcal{D}_i \ddot{L}_k &= L_i \times \ddot{L}_k + \frac{\partial \ddot{L}_k}{\partial q_i} \\ &= [L_i, \ddot{L}_k] + [L_k, \ddot{L}_i] - [L_i, \ddot{L}_k] + 2\,\dot{L}_k \times \dot{L}_i \\ &= L_k \times \ddot{L}_i + 2\,\dot{L}_k \times \dot{L}_i, \end{aligned} \tag{74}$$

which proves (67). Equations (68)–(69) are direct specializations. □

Lemma 2 (Ordinary derivatives with respect to $\dot{q}_i$ and $\ddot{q}_i$)**.** *The following identities hold:*

$$\frac{\partial L_k}{\partial \dot{q}_i} = 0, \qquad \frac{\partial L_k}{\partial \ddot{q}_i} = 0, \tag{75}$$

$$\frac{\partial V_k}{\partial \dot{q}_i} = L_i, \qquad \frac{\partial V_k}{\partial \ddot{q}_i} = 0, \tag{76}$$

$$\frac{\partial \dot{L}_k}{\partial \dot{q}_i} = L_k \times L_i, \qquad \frac{\partial \dot{L}_k}{\partial \ddot{q}_i} = 0, \tag{77}$$

$$\frac{\partial \dot{V}_k}{\partial \dot{q}_i} = V_k \times L_i + 2\dot{L}_i, \qquad \frac{\partial \dot{V}_k}{\partial \ddot{q}_i} = L_i, \tag{78}$$

$$\frac{\partial \ddot{L}_k}{\partial \dot{q}_i} = 2(L_k \times \dot{L}_i - L_i \times \dot{L}_k), \qquad \frac{\partial \ddot{L}_k}{\partial \ddot{q}_i} = L_k \times L_i. \tag{79}$$

Proof. All identities are immediate from (47), (49), (51), (52), (54), and (55), together with

$$[V_k - 2V_i, L_i] = [V_k, L_i] - 2[V_i, L_i] = V_k \times L_i + 2\dot{L}_i,$$

since $\dot{L}_i = [L_i, V_i]$ implies $[V_i, L_i] = -\dot{L}_i$. □

4.4 A useful bilinear operator

Define

$$\mathcal{B}_{k,v}(X) := I_k(v \times X) + I_k(X) \times v + I_k(v) \times X. \tag{80}$$

Lemma 3 (Algebraic symmetry identity)**.** *For any Lie-algebra-valued quantities a, b,*

$$\mathcal{B}_{k,a}(b) - \mathcal{B}_{k,b}(a) = 2I_k(a \times b). \tag{81}$$

Equivalently,

$$\mathcal{B}_{k,b}(a) + 2I_k(a \times b) = \mathcal{B}_{k,a}(b). \tag{82}$$

Proof. By definition,

$$\mathcal{B}_{k,a}(b) = I_k(a \times b) + I_k(b) \times a + I_k(a) \times b,$$

and

$$\mathcal{B}_{k,b}(a) = I_k(b \times a) + I_k(a) \times b + I_k(b) \times a.$$

Subtracting the two expressions gives

$$\mathcal{B}_{k,a}(b) - \mathcal{B}_{k,b}(a) = I_k(a \times b) - I_k(b \times a).$$

Since $b \times a = -(a \times b)$, we obtain

$$\mathcal{B}_{k,a}(b) - \mathcal{B}_{k,b}(a) = 2I_k(a \times b).$$

□

Lemma 4 (Derivative rules for $\mathcal{B}_{k,v}(X)$)**.** *The following identities hold:*

$$\mathcal{D}_j\big(\mathcal{B}_{k,v}(X)\big) = \mathcal{B}_{k,v}(\mathcal{D}_j X) + \mathcal{B}_{k,\mathcal{D}_j v}(X), \tag{83}$$

$$\frac{\partial}{\partial \dot{q}_j}\big(\mathcal{B}_{k,v}(X)\big) = \mathcal{B}_{k,v}\left(\frac{\partial X}{\partial \dot{q}_j}\right) + \mathcal{B}_{k,\frac{\partial v}{\partial \dot{q}_j}}(X), \tag{84}$$

$$\frac{\partial}{\partial \ddot{q}_j}\big(\mathcal{B}_{k,v}(X)\big) = \mathcal{B}_{k,v}\left(\frac{\partial X}{\partial \ddot{q}_j}\right) + \mathcal{B}_{k,\frac{\partial v}{\partial \ddot{q}_j}}(X). \tag{85}$$

Proof. We prove (83); the other two are analogous. Using (80) and (62),

$$\begin{aligned}
\mathcal{D}_j\big(\mathcal{B}_{k,v}(X)\big) &= \mathcal{D}_j\big(I_k(v \times X)\big) + \mathcal{D}_j\big(I_k(X) \times v\big) + \mathcal{D}_j\big(I_k(v) \times X\big) \\
&= I_k\big(\mathcal{D}_j(v \times X)\big) + \mathcal{D}_j(I_k(X)) \times v + I_k(X) \times \mathcal{D}_j v + \mathcal{D}_j(I_k(v)) \times X + I_k(v) \times \mathcal{D}_j X \\
&= I_k\big((\mathcal{D}_j v) \times X + v \times (\mathcal{D}_j X)\big) + I_k(\mathcal{D}_j X) \times v + I_k(X) \times \mathcal{D}_j v \\
&\quad + I_k(\mathcal{D}_j v) \times X + I_k(v) \times \mathcal{D}_j X \\
&= \mathcal{B}_{k,v}(\mathcal{D}_j X) + \mathcal{B}_{k,\mathcal{D}_j v}(X). \qquad (86)
\end{aligned}$$

□

4.5 First-order derivatives of $\dot{\Pi}_k$

Proposition 1 (First-order derivatives of $\dot{\Pi}_k$)**.** *The following identities hold:*

$$\mathcal{D}_i \dot{\Pi}_k = I_k(\ddot{L}_i) + \mathcal{B}_{k,V_k}(\dot{L}_i), \qquad (87)$$

$$\frac{\partial \dot{\Pi}_k}{\partial \dot{q}_i} = \mathcal{B}_{k,V_k}(L_i) + 2I_k(\dot{L}_i), \qquad (88)$$

$$\frac{\partial \dot{\Pi}_k}{\partial \ddot{q}_i} = I_k(L_i). \qquad (89)$$

Proof. Starting from (57),

$$\dot{\Pi}_k = I_k(\dot{V}_k) + [I_k(V_k), V_k].$$

For (87), apply $\mathcal{D}_i$:

$$\begin{aligned}
\mathcal{D}_i \dot{\Pi}_k &= \mathcal{D}_i(I_k(\dot{V}_k)) + \mathcal{D}_i\big([I_k(V_k), V_k]\big) \\
&= I_k(\mathcal{D}_i \dot{V}_k) + [\mathcal{D}_i(I_k(V_k)), V_k] + [I_k(V_k), \mathcal{D}_i V_k] \\
&= I_k(\ddot{L}_i + V_k \times \dot{L}_i) + [I_k(\dot{L}_i), V_k] + [I_k(V_k), \dot{L}_i] \\
&= I_k(\ddot{L}_i) + I_k(V_k \times \dot{L}_i) + I_k(\dot{L}_i) \times V_k + I_k(V_k) \times \dot{L}_i \\
&= I_k(\ddot{L}_i) + \mathcal{B}_{k,V_k}(\dot{L}_i). \qquad (90)
\end{aligned}$$

For (88), differentiate with respect to $\dot{q}_i$:

$$\begin{aligned}
\frac{\partial \dot{\Pi}_k}{\partial \dot{q}_i} &= I_k\left(\frac{\partial \dot{V}_k}{\partial \dot{q}_i}\right) + \left[I_k\left(\frac{\partial V_k}{\partial \dot{q}_i}\right), V_k\right] + \left[I_k(V_k), \frac{\partial V_k}{\partial \dot{q}_i}\right] \\
&= I_k(V_k \times L_i + 2\dot{L}_i) + [I_k(L_i), V_k] + [I_k(V_k), L_i] \\
&= \mathcal{B}_{k,V_k}(L_i) + 2I_k(\dot{L}_i). \qquad (91)
\end{aligned}$$

For (89), differentiate with respect to $\ddot{q}_i$:

$$\begin{aligned}
\frac{\partial \dot{\Pi}_k}{\partial \ddot{q}_i} &= I_k\left(\frac{\partial \dot{V}_k}{\partial \ddot{q}_i}\right) + \left[I_k\left(\frac{\partial V_k}{\partial \ddot{q}_i}\right), V_k\right] + \left[I_k(V_k), \frac{\partial V_k}{\partial \ddot{q}_i}\right] \\
&= I_k(L_i) + 0 + 0 \\
&= I_k(L_i). \qquad (92)
\end{aligned}$$

□

4.6 Second-order derivatives of $\dot{\Pi}_k$

Proposition 2 (Second-order derivatives of $\dot{\Pi}_k$)**.** *The following identities hold:*

$$\mathcal{D}_j\mathcal{D}_i\dot{\Pi}_k = I_k(L_i \times \ddot{L}_j) + \mathcal{B}_{k,V_k}(L_i \times \dot{L}_j) + \mathcal{B}_{k,\dot{L}_i}(\dot{L}_j), \tag{93}$$

$$\mathcal{D}_j\left(\frac{\partial\dot{\Pi}_k}{\partial\dot{q}_i}\right) = \mathcal{B}_{k,L_i}(\dot{L}_j), \tag{94}$$

$$\frac{\partial}{\partial\dot{q}_j}\Big(\mathcal{D}_i\dot{\Pi}_k\Big) = 2I_k(L_i \times \dot{L}_j) + \mathcal{B}_{k,V_k}(L_i \times L_j) + \mathcal{B}_{k,\dot{L}_i}(L_j), \tag{95}$$

$$\frac{\partial^2\dot{\Pi}_k}{\partial\dot{q}_j\,\partial\dot{q}_i} = \mathcal{B}_{k,L_i}(L_j), \tag{96}$$

$$\mathcal{D}_j\left(\frac{\partial\dot{\Pi}_k}{\partial\ddot{q}_i}\right) = 0, \tag{97}$$

$$\frac{\partial}{\partial\ddot{q}_j}\Big(\mathcal{D}_i\dot{\Pi}_k\Big) = I_k(L_i \times L_j), \tag{98}$$

$$\frac{\partial}{\partial\ddot{q}_j}\left(\frac{\partial\dot{\Pi}_k}{\partial\dot{q}_i}\right) = 0, \tag{99}$$

$$\frac{\partial}{\partial\ddot{q}_j}\left(\frac{\partial\dot{\Pi}_k}{\partial\ddot{q}_i}\right) = 0. \tag{100}$$

Proof. Start from (87):

$$\mathcal{D}_i\dot{\Pi}_k = I_k(\ddot{L}_i) + \mathcal{B}_{k,V_k}(\dot{L}_i).$$

Apply $\mathcal{D}_j$:

$$\begin{aligned}\mathcal{D}_j\mathcal{D}_i\dot{\Pi}_k &= \mathcal{D}_j(I_k(\ddot{L}_i)) + \mathcal{D}_j\big(\mathcal{B}_{k,V_k}(\dot{L}_i)\big)\\ &= I_k(\mathcal{D}_j\ddot{L}_i) + \mathcal{B}_{k,V_k}(\mathcal{D}_j\dot{L}_i) + \mathcal{B}_{k,\mathcal{D}_jV_k}(\dot{L}_i)\\ &= I_k(L_i \times \ddot{L}_j + 2\dot{L}_i \times \dot{L}_j) + \mathcal{B}_{k,V_k}(L_i \times \dot{L}_j) + \mathcal{B}_{k,\dot{L}_j}(\dot{L}_i).\end{aligned} \tag{101}$$

Now use (82) with $a = \dot{L}_i$ and $b = \dot{L}_j$:

$$\mathcal{B}_{k,\dot{L}_j}(\dot{L}_i) + 2I_k(\dot{L}_i \times \dot{L}_j) = \mathcal{B}_{k,\dot{L}_i}(\dot{L}_j).$$

Substituting this into (101) gives (93).

Next, start from (88):

$$\frac{\partial\dot{\Pi}_k}{\partial\dot{q}_i} = \mathcal{B}_{k,V_k}(L_i) + 2I_k(\dot{L}_i).$$

Apply $\mathcal{D}_j$:

$$\begin{aligned}\mathcal{D}_j\left(\frac{\partial\dot{\Pi}_k}{\partial\dot{q}_i}\right) &= \mathcal{D}_j\big(\mathcal{B}_{k,V_k}(L_i)\big) + 2\mathcal{D}_j(I_k(\dot{L}_i))\\ &= \mathcal{B}_{k,V_k}(\mathcal{D}_jL_i) + \mathcal{B}_{k,\mathcal{D}_jV_k}(L_i) + 2I_k(\mathcal{D}_j\dot{L}_i)\\ &= 0 + \mathcal{B}_{k,\dot{L}_j}(L_i) + 2I_k(L_i \times \dot{L}_j).\end{aligned} \tag{102}$$

Now use (82) with $a = L_i$ and $b = \dot{L}_j$:

$$\mathcal{B}_{k,\dot{L}_j}(L_i) + 2I_k(L_i \times \dot{L}_j) = \mathcal{B}_{k,L_i}(\dot{L}_j),$$

which proves (94).

For (95), apply $\partial/\partial\dot{q}_j$ to (87):

$$\begin{aligned}\frac{\partial}{\partial\dot{q}_j}\Big(\mathcal{D}_i\dot{\Pi}_k\Big) &= I_k\left(\frac{\partial\ddot{L}_i}{\partial\dot{q}_j}\right) + \frac{\partial}{\partial\dot{q}_j}\big(\mathcal{B}_{k,V_k}(\dot{L}_i)\big)\\ &= I_k\Big(2(L_i\times\dot{L}_j - L_j\times\dot{L}_i)\Big) + \mathcal{B}_{k,V_k}\left(\frac{\partial\dot{L}_i}{\partial\dot{q}_j}\right) + \mathcal{B}_{k,\frac{\partial V_k}{\partial\dot{q}_j}}(\dot{L}_i)\\ &= 2I_k(L_i\times\dot{L}_j - L_j\times\dot{L}_i) + \mathcal{B}_{k,V_k}(L_i\times L_j) + \mathcal{B}_{k,L_j}(\dot{L}_i). \end{aligned} \tag{103}$$

Use (81) with $a=\dot{L}_i$ and $b=L_j$:

$$\mathcal{B}_{k,\dot{L}_i}(L_j) - \mathcal{B}_{k,L_j}(\dot{L}_i) = 2I_k(\dot{L}_i\times L_j) = -2I_k(L_j\times\dot{L}_i).$$

Hence

$$\mathcal{B}_{k,L_j}(\dot{L}_i) - 2I_k(L_j\times\dot{L}_i) = \mathcal{B}_{k,\dot{L}_i}(L_j).$$

Substituting this into (103) proves (95).

For (96), apply $\partial/\partial\dot{q}_j$ to (88):

$$\begin{aligned}\frac{\partial^2\dot{\Pi}_k}{\partial\dot{q}_j\,\partial\dot{q}_i} &= \frac{\partial}{\partial\dot{q}_j}\big(\mathcal{B}_{k,V_k}(L_i)\big) + 2I_k\left(\frac{\partial\dot{L}_i}{\partial\dot{q}_j}\right)\\ &= \mathcal{B}_{k,V_k}\left(\frac{\partial L_i}{\partial\dot{q}_j}\right) + \mathcal{B}_{k,\frac{\partial V_k}{\partial\dot{q}_j}}(L_i) + 2I_k(L_i\times L_j)\\ &= 0 + \mathcal{B}_{k,L_j}(L_i) + 2I_k(L_i\times L_j). \end{aligned} \tag{104}$$

Using (82) with $a=L_i$ and $b=L_j$ yields

$$\mathcal{B}_{k,L_j}(L_i) + 2I_k(L_i\times L_j) = \mathcal{B}_{k,L_i}(L_j),$$

which proves (96).

For (97), start from (89):

$$\mathcal{D}_j\left(\frac{\partial\dot{\Pi}_k}{\partial\ddot{q}_i}\right) = \mathcal{D}_j(I_k(L_i)) = I_k(\mathcal{D}_jL_i) = 0.$$

For (98), apply $\partial/\partial\ddot{q}_j$ to (87):

$$\begin{aligned}\frac{\partial}{\partial\ddot{q}_j}\Big(\mathcal{D}_i\dot{\Pi}_k\Big) &= I_k\left(\frac{\partial\ddot{L}_i}{\partial\ddot{q}_j}\right) + \frac{\partial}{\partial\ddot{q}_j}\big(\mathcal{B}_{k,V_k}(\dot{L}_i)\big)\\ &= I_k(L_i\times L_j) + \mathcal{B}_{k,V_k}(0) + \mathcal{B}_{k,0}(\dot{L}_i)\\ &= I_k(L_i\times L_j). \end{aligned} \tag{105}$$

For (99), differentiate (88) with respect to $\ddot{q}_j$:

$$\frac{\partial}{\partial\ddot{q}_j}\left(\frac{\partial\dot{\Pi}_k}{\partial\dot{q}_i}\right) = \frac{\partial}{\partial\ddot{q}_j}\big(\mathcal{B}_{k,V_k}(L_i)\big) + 2I_k\left(\frac{\partial\dot{L}_i}{\partial\ddot{q}_j}\right) = 0.$$

For (100), differentiate (89) with respect to $\ddot{q}_j$:

$$\frac{\partial}{\partial \ddot{q}_j}\left(\frac{\partial \dot{\Pi}_k}{\partial \ddot{q}_i}\right) = \frac{\partial}{\partial \ddot{q}_j} I_k(L_i) = 0.$$

This completes the proof. □

4.7 First-order derivatives of τ_k

Proposition 3 (First-order derivatives of τ_k)**.** *The following identities hold:*

$$\frac{\partial \tau_k}{\partial q_i} = L_k * \Big(I_k(\ddot{L}_i) + \mathcal{B}_{k,V_k}(\dot{L}_i)\Big), \tag{106}$$

$$\frac{\partial \tau_k}{\partial \dot{q}_i} = L_k * \Big(\mathcal{B}_{k,V_k}(L_i) + 2I_k(\dot{L}_i)\Big), \tag{107}$$

$$\frac{\partial \tau_k}{\partial \ddot{q}_i} = L_k * I_k(L_i). \tag{108}$$

Proof. Using (40),

$$\begin{aligned}\frac{\partial \tau_k}{\partial q_i} &= \frac{\partial}{\partial q_i}(L_k * \dot{\Pi}_k)\\ &= L_k * \left(L_i \times \dot{\Pi}_k + \frac{\partial \dot{\Pi}_k}{\partial q_i}\right)\\ &= L_k * (\mathcal{D}_i \dot{\Pi}_k)\\ &= L_k * \Big(I_k(\ddot{L}_i) + \mathcal{B}_{k,V_k}(\dot{L}_i)\Big),\end{aligned} \tag{109}$$

which proves (106).

Using (41) and (88),

$$\frac{\partial \tau_k}{\partial \dot{q}_i} = L_k * \frac{\partial \dot{\Pi}_k}{\partial \dot{q}_i} = L_k * \Big(\mathcal{B}_{k,V_k}(L_i) + 2I_k(\dot{L}_i)\Big),$$

which proves (107).

Using (42) and (89),

$$\frac{\partial \tau_k}{\partial \ddot{q}_i} = L_k * \frac{\partial \dot{\Pi}_k}{\partial \ddot{q}_i} = L_k * I_k(L_i),$$

which proves (108). □

4.8 Second-order derivatives of τ_k

Proposition 4 (Second-order derivatives of τ_k)**.** *The ordered second derivatives of τ_k are given by*

$$\frac{\partial^2 \tau_k}{\partial q_j\, \partial q_i} = L_k * \Big(I_k(L_i \times \ddot{L}_j) + \mathcal{B}_{k,V_k}(L_i \times \dot{L}_j) + \mathcal{B}_{k,\dot{L}_i}(\dot{L}_j) \Big), \tag{110}$$

$$\frac{\partial^2 \tau_k}{\partial q_j\, \partial \dot{q}_i} = L_k * \mathcal{B}_{k,L_i}(\dot{L}_j), \tag{111}$$

$$\frac{\partial^2 \tau_k}{\partial \dot{q}_j\, \partial q_i} = L_k * \Big(2I_k(L_i \times \dot{L}_j) + \mathcal{B}_{k,V_k}(L_i \times L_j) + \mathcal{B}_{k,\dot{L}_i}(L_j) \Big), \tag{112}$$

$$\frac{\partial^2 \tau_k}{\partial \dot{q}_j\, \partial \dot{q}_i} = L_k * \mathcal{B}_{k,L_i}(L_j), \tag{113}$$

$$\frac{\partial^2 \tau_k}{\partial q_j\, \partial \ddot{q}_i} = 0, \tag{114}$$

$$\frac{\partial^2 \tau_k}{\partial \ddot{q}_j\, \partial q_i} = L_k * I_k(L_i \times L_j), \tag{115}$$

$$\frac{\partial^2 \tau_k}{\partial \dot{q}_j\, \partial \ddot{q}_i} = 0, \tag{116}$$

$$\frac{\partial^2 \tau_k}{\partial \ddot{q}_j\, \partial \dot{q}_i} = 0, \tag{117}$$

$$\frac{\partial^2 \tau_k}{\partial \ddot{q}_j\, \partial \ddot{q}_i} = 0. \tag{118}$$

Proof. We repeatedly use (40)–(42) together with the already established derivatives of $\dot{\Pi}_k$.

1. Derivative $\partial^2 \tau_k / (\partial q_j\, \partial q_i)$. Starting from (106),

$$\begin{aligned} \frac{\partial^2 \tau_k}{\partial q_j\, \partial q_i} &= \frac{\partial}{\partial q_j} \Big(L_k * (\mathcal{D}_i \dot{\Pi}_k) \Big) \\ &= L_k * \Big(\mathcal{D}_j (\mathcal{D}_i \dot{\Pi}_k) \Big) \\ &= L_k * (\mathcal{D}_j \mathcal{D}_i \dot{\Pi}_k). \end{aligned} \tag{119}$$

Substituting (93) gives (110).

2. Derivative $\partial^2 \tau_k / (\partial q_j\, \partial \dot{q}_i)$. Starting from (107),

$$\begin{aligned} \frac{\partial^2 \tau_k}{\partial q_j\, \partial \dot{q}_i} &= \frac{\partial}{\partial q_j} \left(L_k * \frac{\partial \dot{\Pi}_k}{\partial \dot{q}_i} \right) \\ &= L_k * \left(\mathcal{D}_j \left(\frac{\partial \dot{\Pi}_k}{\partial \dot{q}_i} \right) \right). \end{aligned} \tag{120}$$

Substituting (94) gives (111).

3. Derivative $\partial^2\tau_k/(\partial\dot{q}_j\,\partial q_i)$. Starting from (106),

$$\begin{aligned}\frac{\partial^2\tau_k}{\partial\dot{q}_j\,\partial q_i} &= \frac{\partial}{\partial\dot{q}_j}\Big(L_k * (\mathcal{D}_i\dot{\Pi}_k)\Big)\\ &= L_k * \left(\frac{\partial}{\partial\dot{q}_j}(\mathcal{D}_i\dot{\Pi}_k)\right).\end{aligned} \tag{121}$$

Substituting (95) gives (112).

4. Derivative $\partial^2\tau_k/(\partial\dot{q}_j\,\partial\dot{q}_i)$. Starting from (107),

$$\begin{aligned}\frac{\partial^2\tau_k}{\partial\dot{q}_j\,\partial\dot{q}_i} &= \frac{\partial}{\partial\dot{q}_j}\left(L_k * \frac{\partial\dot{\Pi}_k}{\partial\dot{q}_i}\right)\\ &= L_k * \left(\frac{\partial^2\dot{\Pi}_k}{\partial\dot{q}_j\,\partial\dot{q}_i}\right).\end{aligned} \tag{122}$$

Substituting (96) gives (113).

5. Derivative $\partial^2\tau_k/(\partial q_j\,\partial\ddot{q}_i)$. Starting from (108),

$$\begin{aligned}\frac{\partial^2\tau_k}{\partial q_j\,\partial\ddot{q}_i} &= \frac{\partial}{\partial q_j}\left(L_k * I_k(L_i)\right)\\ &= L_k * (\mathcal{D}_j(I_k(L_i)))\\ &= 0,\end{aligned} \tag{123}$$

which proves (114).

6. Derivative $\partial^2\tau_k/(\partial\ddot{q}_j\,\partial q_i)$. Starting from (106),

$$\begin{aligned}\frac{\partial^2\tau_k}{\partial\ddot{q}_j\,\partial q_i} &= \frac{\partial}{\partial\ddot{q}_j}\Big(L_k * (\mathcal{D}_i\dot{\Pi}_k)\Big)\\ &= L_k * \left(\frac{\partial}{\partial\ddot{q}_j}(\mathcal{D}_i\dot{\Pi}_k)\right)\\ &= L_k * I_k(L_i \times L_j),\end{aligned} \tag{124}$$

which proves (115).

7. Derivatives involving $\ddot{q}$ twice or mixed with $\dot{q}$. From (108),

$$\frac{\partial\tau_k}{\partial\ddot{q}_i} = L_k * I_k(L_i),$$

which does not depend on $\dot{q}_j$ or $\ddot{q}_j$. Therefore,

$$\frac{\partial^2\tau_k}{\partial\dot{q}_j\,\partial\ddot{q}_i} = \frac{\partial^2\tau_k}{\partial\ddot{q}_j\,\partial\ddot{q}_i} = 0.$$

Similarly, (99) implies

$$\frac{\partial^2\tau_k}{\partial\ddot{q}_j\,\partial\dot{q}_i} = 0.$$

Table 1: Accuracy validation of first-order analytical inverse-dynamics derivatives on Franka Emika Panda.

Derivative item	Median rel. err.	Worst rel. err.	Median max abs. err.	Worst max abs. err.
$\partial\tau/\partial q$	5.55×10^{-11}	2.93×10^{-10}	1.177×10^{-9}	4.384×10^{-9}
$\partial\tau/\partial \dot{q}$	4.013×10^{-10}	2.207×10^{-9}	9.59×10^{-10}	3.093×10^{-9}
$\partial\tau/\partial \ddot{q}$	5.362×10^{-10}	1.311×10^{-9}	8.94×10^{-10}	3.889×10^{-9}

Table 2: Accuracy validation of second-order analytical inverse-dynamics derivatives on Franka Emika Panda.

Derivative item	Median rel. err.	Worst rel. err.	Median max abs. err.	Worst max abs. err.
$\partial^2\tau/\partial q^2$	4.55×10^{-8}	7.87×10^{-8}	1.353×10^{-6}	3.691×10^{-6}
$\partial^2\tau/\partial q\,\partial \dot{q}$	1.315×10^{-7}	5.233×10^{-7}	3.68×10^{-7}	8.24×10^{-7}
$\partial^2\tau/\partial \dot{q}^2$	4.435×10^{-7}	7.890×10^{-7}	9.92×10^{-7}	2.127×10^{-6}
$\partial^2\tau/\partial q\,\partial \ddot{q}$	3.081×10^{-7}	5.503×10^{-7}	3.30×10^{-7}	6.16×10^{-7}

This proves (116)–(118). □

5 Derivative Correctness

References

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